# AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 9: Singular Value Decomposition (SVD)

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## 1 Keywords

- Singular Value Decomposition, SVD, spectral norm, condition number, general linear maps, ridge regression

## 2 Main Points

### 2.1 Singular Value Decomposition

- Any $m \times p$ matrix, $\mathbf{X}$ can be decomposed as $\mathbf{X} = \mathbf{U}\,\mathbf{S}\,\mathbf{V}^\mathsf{T}$ (for a design matrix, $m$ is the number of data points and $p$ the number of features)
  - $\mathbf{U}$ is an $m \times m$ orthogonal matrix
  - $\mathbf{S}$ is an $m \times p$ matrix with zeros everywhere except the diagonal where $S_{ii} = s_i \geq 0$
  - $\mathbf{V}$ is a $p \times p$ orthogonal matrix
- The values $s_i$ are known as the *singular values* of $\mathbf{X}$
- The columns $\boldsymbol{u}_i$ of $\mathbf{U}$ and $\boldsymbol{v}_i$ of $\mathbf{V}$ are the left and right *singular vectors*. Column by column the SVD says $\mathbf{X}\,\boldsymbol{v}_i = s_i\,\boldsymbol{u}_i$ and $\mathbf{X}^\mathsf{T}\boldsymbol{u}_i = s_i\,\boldsymbol{v}_i$
- If $\mathbf{X}$ is symmetric and positive semi-definite (e.g. a covariance matrix) then $\mathbf{U} = \mathbf{V}$ and $\mathbf{S} = \mathbf{\Lambda}$: the SVD is the eigen-decomposition. For a general symmetric matrix $s_i = |\lambda_i|$, and the columns of $\mathbf{U}$ and $\mathbf{V}$ belonging to a negative eigenvalue differ in sign
- **Where it comes from.** The SVD can be built from the eigenvectors of a symmetric matrix; the derivation is in the next section
- **Economical SVD**
  - If $m > p$ some algorithms won't bother outputting the last $m - p$ columns of $\mathbf{U}$
  - If $p > m$ some algorithms won't bother outputting the last $p - m$ columns of $\mathbf{V}$
  - In this case it will output a square $\mathbf{S}$ matrix
  - In numpy, `U, s, Vt = np.linalg.svd(X)` gives the full form and `np.linalg.svd(X, full_matrices=False)` the economical one. Note that numpy returns the singular values as a vector `s` and returns $\mathbf{V}^\mathsf{T}$, not $\mathbf{V}$
- **Finding the SVD.** In principle we can take whichever of $\mathbf{X}\,\mathbf{X}^\mathsf{T}$ and $\mathbf{X}^\mathsf{T}\mathbf{X}$ is smaller, find its eigenvalues $s_i^2$ and eigenvectors, and recover the other set of singular vectors from $\mathbf{U} = \mathbf{X}\,\mathbf{V}\,\mathbf{S}^{-1}$ (for the non-zero $s_i$). Good libraries do not do this: forming $\mathbf{X}^\mathsf{T}\mathbf{X}$ squares the condition number (see the mappings notes), so they use a more stable algorithm on $\mathbf{X}$ itself

## 2.2 Where the SVD Comes From

- Take $m \le p$ (otherwise apply the argument to $\mathbf{X}^\mathsf{T}$) and build the $(m+p)\times(m+p)$ matrix

$$\mathbf{B} = \begin{pmatrix} \mathbf{0} & \mathbf{X} \\ \mathbf{X}^\mathsf{T} & \mathbf{0} \end{pmatrix}$$

  It is symmetric, so it has real eigenvalues and orthonormal eigenvectors. This is the whole trick: everything we know about symmetric matrices transfers to an arbitrary $\mathbf{X}$

- **Eigenvectors come in pairs.** Write an eigenvector as $(\boldsymbol{u}, \boldsymbol{v})$, with $\boldsymbol{u} \in \mathbb{R}^m$ and $\boldsymbol{v} \in \mathbb{R}^p$. Then $\mathbf{B}\,(\boldsymbol{u}, \boldsymbol{v}) = s\,(\boldsymbol{u}, \boldsymbol{v})$ says exactly

$$\mathbf{X}\,\boldsymbol{v} = s\,\boldsymbol{u}, \qquad \mathbf{X}^\mathsf{T}\boldsymbol{u} = s\,\boldsymbol{v}$$

  Changing the sign of $\boldsymbol{v}$ changes the sign of both right-hand sides, so $(\boldsymbol{u}, -\boldsymbol{v})$ is an eigenvector with eigenvalue $-s$. The eigenvalues therefore come as $m$ pairs $\pm s_i$ (with $s_i \ge 0$) together with $p - m$ zeros, whose eigenvectors have the form $(\mathbf{0}, \boldsymbol{v}_k)$ with $\mathbf{X}\,\boldsymbol{v}_k = \mathbf{0}$

- Applying $\mathbf{X}^\mathsf{T}$ to the first equation gives $\mathbf{X}^\mathsf{T}\mathbf{X}\,\boldsymbol{v} = s\,\mathbf{X}^\mathsf{T}\boldsymbol{u} = s^2\,\boldsymbol{v}$, and similarly $\mathbf{X}\,\mathbf{X}^\mathsf{T}\boldsymbol{u} = s^2\,\boldsymbol{u}$: the $s_i^2$ are the eigenvalues of $\mathbf{X}^\mathsf{T}\mathbf{X}$ and $\mathbf{X}\,\mathbf{X}^\mathsf{T}$

- **Normalisation: why $\|\boldsymbol{u}\| = \|\boldsymbol{v}\|$.** Normalise the eigenvector so that $\|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2 = 1$. If its eigenvalue $s$ is not zero, the two halves have the same length. Take the inner product of $\mathbf{X}\,\boldsymbol{v} = s\,\boldsymbol{u}$ with $\boldsymbol{u}$, and of $\mathbf{X}^\mathsf{T}\boldsymbol{u} = s\,\boldsymbol{v}$ with $\boldsymbol{v}$:

$$s\,\|\boldsymbol{u}\|^2 = \boldsymbol{u}^\mathsf{T}\mathbf{X}\,\boldsymbol{v}, \qquad s\,\|\boldsymbol{v}\|^2 = \boldsymbol{v}^\mathsf{T}\mathbf{X}^\mathsf{T}\boldsymbol{u}$$

  The right-hand sides are equal, since $\boldsymbol{v}^\mathsf{T}\mathbf{X}^\mathsf{T}\boldsymbol{u} = (\boldsymbol{u}^\mathsf{T}\mathbf{X}\,\boldsymbol{v})^\mathsf{T}$ and a number equals its own transpose. So $s\,\|\boldsymbol{u}\|^2 = s\,\|\boldsymbol{v}\|^2$, and dividing by $s \ne 0$,

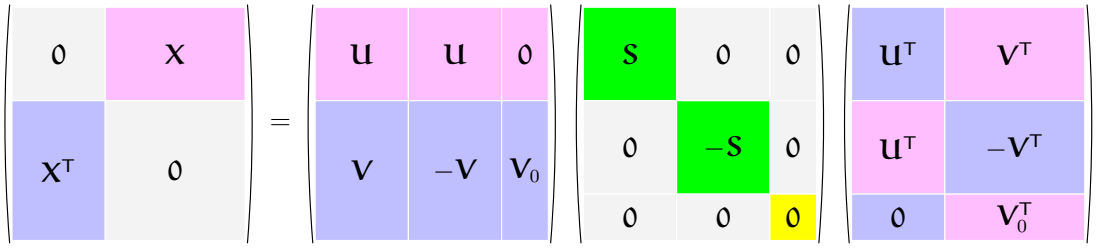
$$\|\boldsymbol{u}\|^2 = \|\boldsymbol{v}\|^2 = \tfrac{1}{2}$$

  - In words: $\mathbf{X}$ takes $\boldsymbol{v}$ to $s\,\boldsymbol{u}$ and $\mathbf{X}^\mathsf{T}$ takes $\boldsymbol{u}$ back to $s\,\boldsymbol{v}$. Both directions scale by the same factor, so neither half can be longer than the other
  - Another route: $(\boldsymbol{u}, \boldsymbol{v})$ and $(\boldsymbol{u}, -\boldsymbol{v})$ have eigenvalues $s$ and $-s$, which differ when $s \ne 0$, so they are orthogonal: $0 = \|\boldsymbol{u}\|^2 - \|\boldsymbol{v}\|^2$
  - It fails when $s = 0$. The equations become $\mathbf{X}\,\boldsymbol{v} = \mathbf{0}$ and $\mathbf{X}^\mathsf{T}\boldsymbol{u} = \mathbf{0}$, which no longer link $\boldsymbol{u}$ to $\boldsymbol{v}$. For example the eigenvectors $(\mathbf{0}, \boldsymbol{v}_k)$ have $\|\boldsymbol{u}\| = 0$ and $\|\boldsymbol{v}\| = 1$
  - Similarly, $(\boldsymbol{u}_i, \boldsymbol{v}_i)$ is orthogonal to both $(\boldsymbol{u}_j, \pm\boldsymbol{v}_j)$ for $j \ne i$, giving $\boldsymbol{u}_i^\mathsf{T}\boldsymbol{u}_j = \boldsymbol{v}_i^\mathsf{T}\boldsymbol{v}_j = 0$. (For simplicity we take every $s_i > 0$)

- **Stacking the eigenvectors.** Put the eigenvectors into the orthogonal matrix $\mathbf{W}$ and the eigenvalues into the diagonal matrix $\mathbf{D}$:

$$\mathbf{W} = \begin{pmatrix} \mathbf{U} & \mathbf{U} & \mathbf{0} \\ \mathbf{V} & -\mathbf{V} & \mathbf{V}_0 \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} \mathbf{S} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{S} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix}$$

  where $\mathbf{U}$ $(m \times m)$ and $\mathbf{V}$ $(p \times m)$ have columns $\boldsymbol{u}_i$ and $\boldsymbol{v}_i$, $\mathbf{V}_0$ $(p \times (p-m))$ has columns $\boldsymbol{v}_k$, and $\mathbf{S} = \mathrm{diag}(s_1, \ldots, s_m)$. The eigen-decomposition of $\mathbf{B}$ is $\mathbf{B} = \mathbf{W}\,\mathbf{D}\,\mathbf{W}^\mathsf{T}$



- **Multiplying out.** First multiply $\mathbf{W}$ by the diagonal matrix $\mathbf{D}$, which scales each block column. Note that the $-\mathbf{V}$ block meets $-\mathbf{S}$, giving $+\mathbf{VS}$, and $\mathbf{V}_0$ meets $\mathbf{0}$

$$\begin{pmatrix} \mathbf{0} & \mathbf{X} \\ \mathbf{X}^\mathsf{T} & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{US} & -\mathbf{US} & \mathbf{0} \\ \mathbf{VS} & \mathbf{VS} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{U}^\mathsf{T} & \mathbf{V}^\mathsf{T} \\ \mathbf{U}^\mathsf{T} & -\mathbf{V}^\mathsf{T} \\ \mathbf{0} & \mathbf{V}_0^\mathsf{T} \end{pmatrix}$$

- Then multiply by $\mathbf{W}^\mathsf{T}$. The top-left block is $\mathbf{USU}^\mathsf{T} - \mathbf{USU}^\mathsf{T} = \mathbf{0}$, the top-right is $\mathbf{USV}^\mathsf{T} + \mathbf{USV}^\mathsf{T} = 2\,\mathbf{USV}^\mathsf{T}$, and similarly for the bottom row

$$\begin{pmatrix} \mathbf{0} & \mathbf{X} \\ \mathbf{X}^\mathsf{T} & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & 2\mathbf{USV}^\mathsf{T} \\ 2\mathbf{VSU}^\mathsf{T} & \mathbf{0} \end{pmatrix}$$

- Comparing the two sides block by block, $\mathbf{X} = 2\,\mathbf{U}\,\mathbf{S}\,\mathbf{V}^\mathsf{T}$. The columns of $\mathbf{U}$ and $\mathbf{V}$ are orthogonal but have length $1/\sqrt{2}$, so define $\tilde{\mathbf{U}} = \sqrt{2}\,\mathbf{U}$ and $\tilde{\mathbf{V}} = \sqrt{2}\,\mathbf{V}$, whose columns are orthonormal. Then

$$\mathbf{X} = \tilde{\mathbf{U}}\,\mathbf{S}\,\tilde{\mathbf{V}}^\mathsf{T}$$

  which is the (economical) SVD. Adding the columns of $\mathbf{V}_0$ to $\tilde{\mathbf{V}}$ completes it to a $p \times p$ orthogonal matrix and gives the full SVD

## 2.3 Spectrum and the Spectral Norm

- The singular values, ordered $s_1 \geq s_2 \geq \cdots \geq 0$, are the (singular value) *spectrum* of $\mathbf{X}$
- The largest, $s_{\max}$, is the most that $\mathbf{X}$ can stretch a vector. It is the *spectral norm* from the vector spaces lecture:

$$\|\mathbf{X}\|_2 = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\mathbf{X}\,\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2} = s_{\max}$$

- **Proof.** Write $\boldsymbol{x} = \sum_i c_i\,\boldsymbol{v}_i$ in the orthonormal basis of right singular vectors, so $\|\boldsymbol{x}\|^2 = \sum_i c_i^2$. Then $\mathbf{X}\,\boldsymbol{x} = \sum_i c_i\,s_i\,\boldsymbol{u}_i$, and as the $\boldsymbol{u}_i$ are orthonormal too

$$\|\mathbf{X}\,\boldsymbol{x}\|^2 = \sum_i s_i^2\,c_i^2 \leq s_{\max}^2 \sum_i c_i^2 = s_{\max}^2\,\|\boldsymbol{x}\|^2$$

  with equality when $\boldsymbol{x} = \boldsymbol{v}_{\max}$, which is mapped to $\mathbf{X}\,\boldsymbol{v}_{\max} = s_{\max}\,\boldsymbol{u}_{\max}$

- Consequences
  - For a symmetric matrix $\|\mathbf{X}\|_2 = \max_i |\lambda_i|$, as used in the eigensystems notes
  - For any square matrix $\|\mathbf{X}\|_2 \geq \max_i |\lambda_i|$ (if $\mathbf{X}\boldsymbol{v} = \lambda\boldsymbol{v}$ then $\boldsymbol{v}$ is stretched by $|\lambda|$, which cannot exceed $s_{\max}$). The two can differ a lot for non-symmetric matrices (Exercise 3)
  - $\|\mathbf{X}\|_F^2 = \sum_{ij} X_{ij}^2 = \sum_i s_i^2 \geq s_{\max}^2$, so $\|\mathbf{X}\|_2 \leq \|\mathbf{X}\|_F$ (Exercise 2)
  - For a square invertible matrix the smallest stretch is $s_{\min}$ and $\left\|\mathbf{X}^{-1}\right\|_2 = 1/s_{\min}$, so the condition number is $\kappa = \|\mathbf{X}\|_2 \left\|\mathbf{X}^{-1}\right\|_2 = s_{\max}/s_{\min}$. This generalises the eigenvalue ratio for symmetric matrices
  - Spectral normalisation of a network layer (vector spaces notes) divides $\mathbf{L}$ by $s_{\max}$ so that $\|\mathbf{L}\|_2 = 1$

## 2.4 General Linear Mapping

- Recall that matrices are the most general linear operators
- Since any matrix $\mathbf{X}$ can be written as $\mathbf{U}\,\mathbf{S}\,\mathbf{V}^\mathsf{T}$ we can interpret any linear mapping as doing three operations
  - A rotation (with possibly a reflection) defined by $\mathbf{V}^\mathsf{T}$
  - A rescaling of each coordinate by $s_i$
  - A rotation (with possibly a reflection) defined by $\mathbf{U}$
- When $\mathbf{X}$ is not square, $\mathbf{S}$ also either squashes some directions to zero or embeds the result in a space with extra dimensions
- The *rank* of any matrix is its number of non-zero singular values (the number of linearly independent rows or columns)
- For a square matrix, using $|\mathbf{AB}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{U}| = \pm 1$ for an orthogonal matrix, $|\det \mathbf{X}| = \prod_i s_i$: the singular values multiply up to the change of volume
- **Duality**
  - Using $\mathbf{X} = \mathbf{U}\,\mathbf{S}\,\mathbf{V}^\mathsf{T}$ then
    - $\mathbf{C} = \mathbf{X}\,\mathbf{X}^\mathsf{T} = \mathbf{U}\mathbf{S}\mathbf{S}^\mathsf{T}\mathbf{U}^\mathsf{T}$
    - $\mathbf{D} = \mathbf{X}^\mathsf{T}\mathbf{X} = \mathbf{V}\mathbf{S}^\mathsf{T}\mathbf{S}\mathbf{V}^\mathsf{T}$
  - $\mathbf{S}\mathbf{S}^\mathsf{T}$ and $\mathbf{S}^\mathsf{T}\mathbf{S}$ are diagonal matrices whose non-zero diagonal elements are $s_i^2$

## 2.5 Ridge Regression

- Ridge regression is linear regression with an $L_2$ regulariser
- Adding a regulariser $\eta\,\|\boldsymbol{w}\|^2$ to the squared loss, the weights, $\boldsymbol{w}^*$, that minimise the loss function are given by $\boldsymbol{w}^* = (\mathbf{X}^\mathsf{T}\mathbf{X} + \eta\,\mathbf{I})^{-1}\mathbf{X}^\mathsf{T}\boldsymbol{y}$
- Using $\mathbf{X} = \mathbf{U}\,\mathbf{S}\,\mathbf{V}^\mathsf{T}$ then

$$\boldsymbol{w}^* = \mathbf{V}\,\hat{\mathbf{S}}^+\mathbf{U}^\mathsf{T}\boldsymbol{y}$$

  where $\hat{\mathbf{S}}^+$ is a regularised pseudo-inverse of $\mathbf{S}$ given by

$$\hat{\mathbf{S}}^+ = (\mathbf{S}^\mathsf{T}\mathbf{S} + \eta\,\mathbf{I})^{-1}\mathbf{S}^\mathsf{T}$$

  - If $\eta = 0$ (and all $s_i > 0$) this is equal to the pseudo-inverse of $\mathbf{S}$
- $\hat{\mathbf{S}}^+$ is a $p \times m$ matrix which is zero everywhere except on the diagonal, where $\hat{S}^+_{ii} = \frac{s_i}{s_i^2 + \eta}$
  - Note if $s_i = 0$ linear regression has an infinity of solutions and the pseudo-inverse of $\mathbf{X}$, as we derived it, does not exist (setting $\eta = 0$ we get $S^+_{ii} = 1/s_i$ which is not defined when $s_i = 0$). See the Moore–Penrose pseudo-inverse below
  - In the regularised case $\hat{S}^+_{ii} = 0$ (we have selected one of the solutions that minimise the squared error)
  - If $s_i \ll \eta$ then without the regularisation term the inverse is very ill-conditioned while with the regularisation term $\hat{S}^+_{ii}$ will be small
  - If $s_i \gg \eta$ then $\hat{S}^+_{ii} \approx \frac{1}{s_i} = S^+_{ii}$
- Adding an $L_2$ regulariser means that the optimum weights, $\boldsymbol{w}^*$, will be less sensitive to the training data reducing the variance in the bias-variance dilemma

### 2.6 The Moore–Penrose Pseudo-Inverse (going further)

- The pseudo-inverse we derived, $\mathbf{X}^{+} = (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}$, only exists when $\mathbf{X}^{\mathsf{T}}\mathbf{X}$ is invertible. That needs every one of the $p$ singular values to be non-zero, which is impossible when $m < p$ (the under-constrained case)
- Using the SVD it can be written $\mathbf{X}^{+} = \mathbf{V}\,\mathbf{S}^{+}\mathbf{U}^{\mathsf{T}}$, where $\mathbf{S}^{+}$ is the $p \times m$ diagonal matrix with $S^{+}_{ii} = 1/s_i$
- The *Moore–Penrose pseudo-inverse* makes one change: wherever $s_i = 0$ it sets $S^{+}_{ii} = 0$ instead of $1/s_i$

  $$S^{+}_{ii} = \begin{cases} 1/s_i & s_i \neq 0 \\ 0 & s_i = 0 \end{cases}$$

  This always exists. When $\mathbf{X}^{\mathsf{T}}\mathbf{X}$ is invertible it is the same as our pseudo-inverse, and when $\mathbf{X}$ is square and invertible it is $\mathbf{X}^{-1}$
- $\boldsymbol{w} = \mathbf{X}^{+}\boldsymbol{y}$ minimises $\|\mathbf{X}\,\boldsymbol{w} - \boldsymbol{y}\|^2$ and, of all the weight vectors that do so, it is the one with the smallest $\|\boldsymbol{w}\|$
- To see why, rotate to the coordinates given by the SVD: let $\boldsymbol{z} = \mathbf{V}^{\mathsf{T}}\boldsymbol{w}$ and $\boldsymbol{c} = \mathbf{U}^{\mathsf{T}}\boldsymbol{y}$. Orthogonal matrices do not change lengths, so $\|\boldsymbol{w}\| = \|\boldsymbol{z}\|$ and

  $$\|\mathbf{X}\,\boldsymbol{w} - \boldsymbol{y}\|^2 = \left\|\mathbf{U}^{\mathsf{T}}(\mathbf{U}\,\mathbf{S}\,\mathbf{V}^{\mathsf{T}}\boldsymbol{w} - \boldsymbol{y})\right\|^2 = \|\mathbf{S}\,\boldsymbol{z} - \boldsymbol{c}\|^2 = \sum_{i:\, s_i \neq 0} (s_i\, z_i - c_i)^2 + \sum_{i:\, s_i = 0} c_i^2$$

  (in the second sum $i$ runs over the rows of $\mathbf{S}$ that have no non-zero singular value)
  - The loss is minimised by $z_i = c_i / s_i$ whenever $s_i \neq 0$
  - The other components of $\boldsymbol{z}$ do not appear in the loss at all: these are the directions $\boldsymbol{v}_i$ with $\mathbf{X}\,\boldsymbol{v}_i = \mathbf{0}$, where the data give no information
  - Every choice of them fits equally well. Setting them to zero gives the smallest $\|\boldsymbol{z}\| = \|\boldsymbol{w}\|$, and that is exactly what $S^{+}_{ii} = 0$ does
- It is also the limit of ridge regression as the regularisation vanishes: as $\eta \to 0$, $\hat{S}^{+}_{ii} = s_i/(s_i^2 + \eta)$ tends to $1/s_i$ if $s_i \neq 0$ and is $0$ if $s_i = 0$
- This is what `np.linalg.pinv` computes, and `np.linalg.lstsq` returns the same minimum-norm solution. In practice "zero" means below a small tolerance, so tiny singular values caused by rounding errors are discarded rather than inverted

## 3 Exercises

### 3.1 Ridge regression

- Ridge regression is just linear regression with an $L_2$ regulariser
  1. Derive the optimal weights in ridge regression
  2. Show that using $\mathbf{X} = \mathbf{U}\,\mathbf{S}\,\mathbf{V}^{\mathsf{T}}$ then $\boldsymbol{w}^{*} = \mathbf{V}\,(\mathbf{S}^{\mathsf{T}}\mathbf{S} + \eta\,\mathbf{I})^{-1}\mathbf{S}^{\mathsf{T}}\,\mathbf{U}^{\mathsf{T}}\boldsymbol{y}$
- See answers

### 3.2 Spectral norm

1. Show that $\|\mathbf{X}\,\boldsymbol{x}\|_2 \le s_{\max}\,\|\boldsymbol{x}\|_2$, with equality for $\boldsymbol{x} = \boldsymbol{v}_{\max}$
2. Show that $\|\mathbf{X}\|_F^2 = \operatorname{Tr}\mathbf{X}^\mathsf{T}\mathbf{X} = \sum_i s_i^2$, and hence that $\|\mathbf{X}\|_2 \le \|\mathbf{X}\|_F$
3. For $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$
   (a) Show that both eigenvalues of $\mathbf{J}$ are $1$
   (b) Find the singular values of $\mathbf{J}$ from the eigenvalues of $\mathbf{J}^\mathsf{T}\mathbf{J}$, and hence $\|\mathbf{J}\|_2$ and the condition number
   (c) Repeat for the symmetric matrix $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, whose eigenvalues are $3$ and $-1$

## 4 Experiments

### 4.1 SVD

- Check the claims in these notes on a random $3 \times 4$ matrix

```
import numpy as np

rng = np.random.default_rng(0)
X = rng.normal(size=(3, 4))
U, s, Vt = np.linalg.svd(X)       # s is a vector, Vt is V transposed
S = np.zeros_like(X)
np.fill_diagonal(S, s)            # the 3 x 4 matrix S
V = Vt.T
print(np.allclose(U @ S @ Vt, X))                        # X = U S V^T
print(np.allclose(U @ U.T, np.eye(3)), np.allclose(V.T @ V, np.eye(4)))
print(np.linalg.eigvalsh(X @ X.T)[::-1], s**2)           # eigenvalues are s_i^2
print(np.linalg.eigvalsh(X.T @ X)[::-1])                 # the same, plus a 0

# spectral norm: the largest stretch of any unit vector
print(np.linalg.norm(X, 2), s[0])                        # both s_max
print(np.linalg.norm(X @ V[:, 0]), np.linalg.norm(X @ V[:, 1]))  # s_1, s_2
x = rng.normal(size=4)
print(np.linalg.norm(X @ x) <= s[0] * np.linalg.norm(x))   # never above s_max
print(np.linalg.norm(X, 'fro')**2, np.sum(s**2))           # Frobenius^2

# ridge regression through the SVD
eta = 0.1
lhs = np.linalg.inv(X.T @ X + eta * np.eye(4))
rhs = V @ np.linalg.inv(S.T @ S + eta * np.eye(4)) @ V.T
print(np.allclose(lhs, rhs))
print(np.linalg.cond(X.T @ X))    # X^T X is singular (rank 3): enormous
```

- Every check prints `True`. The unit vector $\boldsymbol{v}_1$ is stretched by exactly $s_1$ and $\boldsymbol{v}_2$ by $s_2$
- $\mathbf{X}^\mathsf{T}\mathbf{X}$ is $4 \times 4$ but only has rank $3$, so it has no inverse. Rounding errors mean numpy may still return one, with a condition number around $10^{16}$: a warning sign never to ignore. Adding $\eta\,\mathbf{I}$ makes it safely invertible

# 5 Answers

## 5.1 Ridge regression

1. It is straightforward to show
$$\boldsymbol{w}^* = (\mathbf{X}^\mathsf{T}\mathbf{X} + \eta\,\mathbf{I})^{-1}\mathbf{X}^\mathsf{T}\boldsymbol{y}$$

2. The only hard part is to show that
$$(\mathbf{X}^\mathsf{T}\mathbf{X} + \eta\,\mathbf{I})^{-1} = \mathbf{V}\,(\mathbf{S}^\mathsf{T}\mathbf{S} + \eta\,\mathbf{I})^{-1}\mathbf{V}^\mathsf{T}$$
   - It is easy to show that $\mathbf{X}^\mathsf{T}\mathbf{X} = \mathbf{V}\,\mathbf{S}^\mathsf{T}\mathbf{S}\,\mathbf{V}^\mathsf{T}$
   - But we also have $\mathbf{I} = \mathbf{V}\,\mathbf{V}^\mathsf{T}$ as $\mathbf{V}$ is an orthogonal matrix
   - Thus $\mathbf{M} = \mathbf{X}^\mathsf{T}\mathbf{X} + \eta\,\mathbf{I} = \mathbf{V}\,(\mathbf{S}^\mathsf{T}\mathbf{S} + \eta\mathbf{I})\mathbf{V}^\mathsf{T} = \mathbf{V}\,\mathbf{W}\,\mathbf{V}^\mathsf{T}$ where $\mathbf{W} = \mathbf{S}^\mathsf{T}\mathbf{S} + \eta\mathbf{I}$
   - But $(\mathbf{A}\,\mathbf{B}\,\mathbf{C})^{-1} = \mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$ (which we can verify by multiplying $\mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$ on either the left or right by $\mathbf{A}\,\mathbf{B}\,\mathbf{C}$)
   - Thus $\mathbf{M}^{-1} = (\mathbf{V}\,\mathbf{W}\,\mathbf{V}^\mathsf{T})^{-1} = (\mathbf{V}^\mathsf{T})^{-1}\,\mathbf{W}^{-1}\mathbf{V}^{-1} = \mathbf{V}\,\mathbf{W}^{-1}\mathbf{V}^\mathsf{T}$ where we use $\mathbf{V}^{-1} = \mathbf{V}^\mathsf{T}$ as $\mathbf{V}$ is an orthogonal matrix

## 5.2 Spectral norm

1. See the proof in the main points: expand $\boldsymbol{x}$ in the right singular vectors, so $\|\mathbf{X}\boldsymbol{x}\|^2 = \sum_i s_i^2 c_i^2 \leq s_{\max}^2 \sum_i c_i^2$. For $\boldsymbol{x} = \boldsymbol{v}_{\max}$ only one $c_i$ is non-zero and the inequality is an equality

2. $\sum_{ij} X_{ij}^2 = \sum_j (\mathbf{X}^\mathsf{T}\mathbf{X})_{jj} = \operatorname{Tr}\mathbf{X}^\mathsf{T}\mathbf{X}$. With $\mathbf{X}^\mathsf{T}\mathbf{X} = \mathbf{V}\,\mathbf{S}^\mathsf{T}\mathbf{S}\,\mathbf{V}^\mathsf{T}$ and the cyclic property of the trace this is $\operatorname{Tr}\mathbf{S}^\mathsf{T}\mathbf{S} = \sum_i s_i^2$. Every term is non-negative, so the sum is at least $s_{\max}^2$

3. Two $2 \times 2$ matrices

   (a) $\mathbf{J}$ is triangular, so its eigenvalues are its diagonal entries, 1 and 1

   (b) $\mathbf{J}^\mathsf{T}\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ has $\lambda^2 - 3\lambda + 1 = 0$, so $\lambda = (3 \pm \sqrt{5})/2$ and the singular values are $s = (\sqrt{5} \pm 1)/2$, i.e. $1.618$ and $0.618$. So $\|\mathbf{J}\|_2 = (1 + \sqrt{5})/2 \approx 1.618$ (the golden ratio) although every eigenvalue is 1, and $\kappa = 1.618/0.618 \approx 2.62$. The eigenvalues of a non-symmetric matrix do not tell you how much it stretches

   (c) Here $s_i = |\lambda_i| = 3$ and 1, so the spectral norm is 3 and $\kappa = 3$. For a symmetric matrix the eigenvalues do tell the whole story